

Numerical Solution for the Hydrogen Atom Ground State

Finite-Difference Method in Scilab/MATLAB

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1. Clear the Environment

```
clear;  
clc;  
clf;
```

- `clear;`: Removes all variables stored in memory. Prevents old variables from affecting the new calculation.
- `clc;`: Clears the console window. Makes the output easier to read.
- `clf;`: Clears any previously drawn figure. Ensures a fresh graph is plotted.

2. Define Physical Constants

```
hbar = 1.054571817e-34;
```

Reduced Planck constant: $\hbar = \frac{h}{2\pi}$. Unit: J · s. Used in the kinetic energy operator.

```
me = 9.10938356e-31;
```

Mass of electron. Unit: kg.

```
echarge = 1.602176634e-19;
```

Elementary charge: $e = 1.602 \times 10^{-19}$ C.

```
eps0 = 8.854187817e-12;
```

Vacuum permittivity: ϵ_0 . Used in Coulomb's law.

3. Coulomb Constant

```
k = echarge^2 / (4 * %pi * eps0);
```

Computes $k = \frac{e^2}{4\pi\epsilon_0}$. Notice this is not the usual Coulomb constant $\frac{1}{4\pi\epsilon_0}$. Instead, it includes the charge squared because the hydrogen potential is $V(r) = -\frac{e^2}{4\pi\epsilon_0 r}$.

```
J_to_eV = 1.0 / echarge;
```

Conversion factor. Since $1 \text{ eV} = 1.602 \times 10^{-19} \text{ J}$, therefore $E(\text{eV}) = \frac{E(\text{J})}{e}$.

4. Numerical Grid

```
rmax = 1e-9;
```

Maximum radius: $r_{\max} = 10^{-9}$ m. Beyond this distance, the wavefunction is almost zero.

```
N = 2000;
```

Number of grid points. The larger the value, the higher the accuracy but the slower the computation.

```
dr = rmax / N;
```

Grid spacing: $\Delta r = \frac{r_{\max}}{N}$.

```
r = linspace(dr, rmax, N)';
```

Creates a column vector:

$$r = \begin{bmatrix} \Delta r \\ 2\Delta r \\ 3\Delta r \\ \vdots \\ r_{\max} \end{bmatrix}$$

Notice it starts from Δr instead of zero. Why? Because dividing by zero (for the $\frac{1}{r}$ potential) is undefined.

5. Construct the Laplacian Matrix

Why Do We Need the Laplacian Matrix?

The time-independent Schrödinger equation for the hydrogen atom is

$$-\frac{\hbar^2}{2m} \frac{d^2\psi(r)}{dr^2} + V(r)\psi(r) = E\psi(r), \quad (1)$$

where

- $\hbar$ is the reduced Planck constant,
- m is the mass of the electron,
- $V(r)$ is the potential energy,
- $\psi(r)$ is the wavefunction,
- E is the energy eigenvalue.

The term

$$\frac{d^2\psi(r)}{dr^2} \quad (2)$$

represents the **second spatial derivative** of the wavefunction. In one-dimensional problems, this operator is known as the **Laplacian operator**.

Since a computer can only perform calculations at a finite number of discrete points, it cannot evaluate derivatives directly. Therefore, the second derivative is replaced by the finite-difference approximation

$$\frac{d^2\psi}{dr^2} \approx \frac{\psi_{i+1} - 2\psi_i + \psi_{i-1}}{(\Delta r)^2}, \quad (3)$$

where

- ψ_i is the wavefunction at the i^{th} grid point,
- ψ_{i-1} is the wavefunction at the previous grid point,
- ψ_{i+1} is the wavefunction at the next grid point,
- Δr is the spacing between adjacent grid points.

Why Convert It into a Matrix?

The finite-difference equation must be applied at every grid point. Writing the finite-difference approximation for all grid points simultaneously leads to a matrix representation of the second-derivative operator:

$$\frac{d^2}{dr^2} \approx \frac{1}{(\Delta r)^2} \begin{bmatrix} -2 & 1 & 0 & \cdots & 0 \\ 1 & -2 & 1 & \cdots & 0 \\ 0 & 1 & -2 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & 1 & -2 \end{bmatrix}. \quad (4)$$

This matrix is known as the **Laplacian matrix**. It provides a numerical representation of the second derivative and is used to construct the kinetic energy operator,

$$\hat{T} = -\frac{\hbar^2}{2m} \frac{d^2}{dr^2}, \quad (5)$$

which is then combined with the potential energy matrix to form the Hamiltonian matrix,

$$\hat{H} = \hat{T} + \hat{V}. \quad (6)$$

Finally, the Hamiltonian matrix is diagonalized to solve the eigenvalue equation

$$\hat{H}\Psi = E\Psi, \quad (7)$$

from which the energy eigenvalues and corresponding wavefunctions of the hydrogen atom are obtained.

So to create **Laplacian matrix**, we proceed as

```
diag_main = -2 * ones(N,1);
```

Creates $[-2, -2, -2, \dots, -2]^T$. This is the main diagonal.

```
diag_off = ones(N-1,1);
```

Creates $[1, 1, 1, \dots, 1]^T$. These are the upper and lower diagonals.

```
Lap = (diag(diag_main) + diag(diag_off,1) + diag(diag_off,-1)) /
      (dr^2);
```

Constructs the tridiagonal matrix:

$$\frac{1}{\Delta r^2} \begin{bmatrix} -2 & 1 & 0 & 0 \\ 1 & -2 & 1 & 0 \\ 0 & 1 & -2 & 1 \\ 0 & 0 & 1 & -2 \end{bmatrix}$$

This is the finite-difference approximation of $\frac{d^2}{dr^2}$ because:

$$\frac{d^2\psi}{dr^2} \approx \frac{\psi_{i+1} - 2\psi_i + \psi_{i-1}}{\Delta r^2}$$

6. Kinetic Energy Matrix

```
T = -(hbar^2 / (2 * me)) * Lap;
```

Implements $\hat{T} = -\frac{\hbar^2}{2m_e} \frac{d^2}{dr^2}$, which is the kinetic energy operator.

7. Potential Energy Matrix

The potential energy of an electron in a hydrogen atom arises due to the electrostatic attraction between the negatively charged electron and the positively charged proton. According to Coulomb's law, the potential energy is given by

$$V(r) = -\frac{e^2}{4\pi\epsilon_0 r}, \quad (8)$$

where

- e is the elementary charge,
- ϵ_0 is the permittivity of free space,
- r is the radial distance of the electron from the nucleus.

The negative sign indicates that the force between the electron and the proton is attractive. As the electron moves farther from the nucleus, the potential energy approaches zero.

In the Scilab program, the constant

$$k = \frac{e^2}{4\pi\epsilon_0} \quad (9)$$

is first calculated. Therefore, the Coulomb potential can be written as

$$V(r) = -\frac{k}{r}. \quad (10)$$

The corresponding Scilab statement is

```
V = diag(-k ./ r);
```

Here, the operator `./` performs element-wise division. Since `r` is a vector containing the radial coordinates,

$$r = \begin{bmatrix} r_1 \\ r_2 \\ r_3 \\ \vdots \\ r_N \end{bmatrix}, \quad (11)$$

the expression

```
-k ./ r
```

produces the vector

$$\begin{bmatrix} -\frac{k}{r_1} \\ -\frac{k}{r_2} \\ -\frac{k}{r_3} \\ \vdots \\ -\frac{k}{r_N} \end{bmatrix}. \quad (12)$$

The `diag()` function converts this vector into a diagonal matrix,

$$V = \begin{bmatrix} -\frac{k}{r_1} & 0 & 0 & \cdots & 0 \\ 0 & -\frac{k}{r_2} & 0 & \cdots & 0 \\ 0 & 0 & -\frac{k}{r_3} & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & -\frac{k}{r_N} \end{bmatrix}. \quad (13)$$

Only the diagonal elements are nonzero because the Coulomb potential depends only on the position of the electron at each grid point. There is no direct coupling between different grid points through the potential energy operator. The off-diagonal elements are therefore zero.

This potential energy matrix is then added to the kinetic energy matrix to form the Hamiltonian matrix,

$$H = T + V, \quad (14)$$

which is used to solve the Schrödinger equation numerically and obtain the energy eigenvalues and corresponding wavefunctions of the hydrogen atom. Only diagonal elements are nonzero because the potential depends only on position.

8. Hamiltonian Matrix

```
H = T + V;
```

Constructs $H = T + V$, which represents the Hamiltonian operator for the hydrogen atom (for $l = 0$, the s-state):

$$\hat{H} = -\frac{\hbar^2}{2m_e} \frac{d^2}{dr^2} - \frac{e^2}{4\pi\epsilon_0 r}$$

9. Solve the Eigenvalue Problem

```
[Psi, E] = spec(H);
```

Computes $H\Psi = E\Psi$, where:

- **Psi** contains the eigenvectors (wavefunctions).
- **E** is a diagonal matrix of eigenvalues (energies).
- The `spec()` function computes the eigenvalues and eigenvectors of a square matrix.

10. Sort the Eigenvalues

```
E_sorted = gsort(diag(E), "g", "i");
```

Steps:

- `diag(E)` extracts the diagonal elements (the eigenvalues).
- `gsort(..., "g", "i")` sorts them in increasing order (smallest energy first), so the most negative energy (ground state) comes first.

11. Ground-State Energy

```
E0_J = E_sorted(1);  
E0_eV = E0_J * J_to_eV;
```

Stores the lowest eigenvalue (numerical ground-state energy in Joules) and converts the energy to electronvolts.

12. Display the Result

```
disp("Ground state energy of Hydrogen atom (approx): " + string(  
    E0_eV) + " eV");  
disp("Expected exact value: -13.6 eV");
```

Displays the computed ground-state energy (e.g., -13.57 eV if the grid is fine enough). Prints the analytical value for comparison. The exact ground-state energy of hydrogen is $E_1 = -13.6$ eV.

13. Plot the Wavefunction

```
plot(r, Psi(:,2));  
xtitle("H atom Ground-State Wave Function");  
xlabel("r (m)");  
ylabel("Wavefunction");  
xgrid(1,1,1);
```

Plots one of the computed eigenvectors against the radial coordinate.

- `r` is the radial distance.
- `Psi(:,2)` selects the second column. Depending on how `spec()` orders the eigenvectors, the ground state may instead be in another column. Ideally, eigenvectors should be reordered consistently with the sorted eigenvalues before plotting.